

SPILLTRACE

User Guide

What every screen shows, how to use it, and who can do what — written for someone opening the system for the first time.

Covers the console at **<http://localhost:3000>**

Every page in this guide was read from the running application.

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In a hurry? Read section 2 to get in, then open any case and press the **Walkthrough** tab — that one screen explains the whole product using the real numbers from that investigation.

1 · What this system does

Ships sometimes wash out their tanks at sea and leave a slick of oil behind. A radar satellite can see that slick from space, at night and through cloud. SPILLTRACE finds it, checks it is really oil, works out where it started, and lists the ships that were there.

The chain, in order

Step	Machine name	What it actually does
Find a satellite pass	<code>scene.search</code>	Ask the European Copernicus catalogue which Sentinel-1 radar images cover this water in the chosen dates.
Fetch the image	<code>scene.download</code>	Download it and record its checksum, so the exact file behind a result can be identified later.
Clean up the radar image	<code>sar.preprocess</code>	Rescale it so that "dark" means the same thing everywhere in the image.
Look for oil	<code>ml.detect</code>	A trained neural network scores every pixel 0–1 for how oil-like it looks; dark shapes become outlines.
Rule out the look-alikes	<code>detect.verify</code>	Seven physical checks against wind and shape, because low wind and algae look identical to oil.
Get the wind and current	<code>env.fetch</code>	Measured weather for that exact patch of sea at that exact hour, from Copernicus Marine.
Run the sea backwards	<code>drift.hindcast</code>	Push thousands of particles back through the real wind and current to find where the oil started.
Collect ship broadcasts	<code>ais.ingest</code>	Gather the public AIS radio broadcasts for that area and window.
Throw out bad ship data	<code>ais.clean</code>	Flag impossible positions and speeds, each with the reason it was rejected.
Draw each ship's path	<code>traj.build</code>	Join scattered positions into tracks, marking every stretch where reporting stopped.
Find who was there	<code>correlate</code>	Keep only the ships inside both the origin region and the discharge window.
Weigh the evidence	<code>score</code>	Score each remaining ship on six separate factors, then combine them with fixed weights.

The rule that governs everything. Nothing here establishes who is responsible for a spill. A high score means “ask *this ship first*”, not “*this ship did it*”. The nearest ship is not automatically the source, a gap in a ship's radio reporting is never treated as wrongdoing, and a score is not a legal probability. Every screen that shows a score says so in writing.

Two things you will see everywhere

Thing	What it means
REAL / SYNTHETIC / MIXED badge	Where the data came from. REAL is live satellite, ocean and ship data. SYNTHETIC is generated for demonstration. MIXED is both — for example a real trained model run over a generated image. It is always written as a word, never colour alone.
LOW / MODERATE / HIGH band	How strong the evidence is for a candidate ship. The server decides it, and it gets held down to MODERATE whenever the evidence cannot actually separate the top candidates. So a score of 0.94 can still read MODERATE — and when it does, the reason is printed beside it.

2 · Signing in, and who can do what

Open `http://localhost:3000` and sign in. There are two roles. Your current role is always shown in the **top-right corner** of the screen, next to your name.

Accounts on this machine

Email	Password	Role
<code>admin@spilltrace.example.com</code>	<code>SpillAdmin!2026</code>	Administrator
<code>ui-review@spilltrace.in</code>	<code>ReviewPass!2026</code>	Administrator
<code>ui-review-analyst@spilltrace.in</code>	<code>ReviewPass!2026</code>	Analyst

These are development accounts on the local database only. Passwords must be at least 12 characters.

What each role can do

Task	Analyst	Administrator
Open and read cases	Yes	Yes
Create a new case and run the pipeline	Yes	Yes
Read spill, drift, ranking and the evidence report	Yes	Yes
Vessel registry, activity feed, compare, analytics	Yes	Yes
See which data sources are real or synthetic	Yes	Yes
ML Ops — model versions and their measured accuracy	No	Yes
System status — component health, job statistics	No	Yes

The analyst does the investigating; the administrator looks after the machine. An analyst opens cases, reads evidence and decides what is worth following up. An administrator additionally checks that the database, queue and storage are healthy, which detection model is live, and whether any pipeline step is failing. Knowing whether data is synthetic is *not* an administrative privilege — every signed-in user can see that, because it changes how the result should be read.

"I cannot open the admin pages." Check the top-right corner. If it says ANALYST, the pages are not missing — your account simply does not have them. Sign out and sign in with an administrator account from the table above. An analyst who types the address directly gets a clear "Administrator access required" message rather than a broken page.

The sidebar, grouped

Group	Screens
Overview	Dashboard · Situational map · Analytics

Investigations	Cases · New case · Compare cases
Intelligence	Vessel registry · Activity feed
System	ML Ops ADMIN · System status ADMIN · How to read this

Two shortcuts worth knowing. `Ctrl + K` opens a search box from any screen — type a case name or a screen name and jump straight there. And on every screen there is a blue **"What am I looking at?"** panel explaining that screen in plain words; press **Hide** once and it collapses everywhere until you bring it back.

3 · The overview screens

Dashboard /dashboard **EVERYONE**

Where you land after signing in. The overview of everything you have running.

WHAT IS ON IT

- **Four counters** — Investigations, Slicks detected, Candidate vessels ranked, and HIGH-band investigative signals.
- **Recent investigations** — your latest cases with their status, data badge, slick count and candidate count. Click any row to open it.
- **Cases opened per week** and **Evidence strength across candidates** — with a **Chart / Table** toggle if you prefer numbers.
- **Look-alike verification** and **Data sources** — how many detections survived the checks, and which feeds are live.

HOW TO USE IT

Read the four counters for scale, then click the newest case in "Recent investigations". If a figure is unknown it shows as a dash — never as zero, because "we could not count this" and "there were none" are different facts.

Situational map /map **EVERYONE**

Every investigation drawn on one chart of the sea.

WHAT IS ON IT

- Each **blue box** is an area someone is investigating. Each **yellow dot** is the centre of a slick detected there.
- **Status filters** at the top of the side panel: All, Running, Completed, Failed, Draft — each with a live count.
- **Fit all** zooms out to show every case at once.
- A **basemap switcher** (top-right of the map): Satellite, Ocean, Dark, Offline — plus a **Globe view** toggle.

HOW TO USE IT

Click a box, a dot or a row in the list to select that case, then use the arrow to open it.

Careful: a yellow dot marks where the oil *was seen*, which is not where it came from. Where it came from is the purple region on each case's own map.

Analytics /analytics **EVERYONE**

Patterns across all your cases at once, rather than inside one.

WHAT IS ON IT — FOUR NUMBERED SECTIONS

1. **Throughput** — cases opened per week, slicks detected per week.

2. **Look-alikes and evidence strength** — verification outcomes, and how bands are spread across candidates.
3. **Investigative scores** — the score distribution, and mean score per factor.
4. **Vessels and provenance** — which ships are most often ranked, the data mix, and model accuracy over time.

HOW TO USE IT

Pick a time range — **All time**, **Last 30 days** or **Last 90 days** — and every chart below follows it. Each chart has a **Chart / Table** toggle.

Careful: every figure is counted from your own cases. It is not a survey of Indian waters.

4 · Registering a new case, step by step

A case is one patch of sea plus one stretch of time. Everything the system finds is limited to those two things. Open **New case** in the sidebar, or press **Ctrl + K** and type "new case".

- 1 **Draw the area on the map.** Choose **Rectangle** for a quick box, or **Polygon** to click corner by corner around an irregular coastline. With Polygon, double-click or press **Enter** to close the shape; **Undo corner** removes the last point and **Clear** starts over.
- 2 **Or type the bounds instead.** Under "Area of interest" there are four boxes — **West, South, East, North** — in decimal degrees. Fill them in and press **Apply bounds**. This is the reliable way to reproduce an exact area.
- 3 **Give it a title.** Required. Write something you will recognise in the case list, for example "Gulf of Kutch — suspected operational discharge".
- 4 **Add a description.** Optional, but it is context for whoever reads the evidence report later.
- 5 **Set the time window — the easy way.** Press one of the **Quick window** buttons: **Last 24 hours**, **Last 7 days** or **Last 30 days**. Both fields fill in at once. This is the dependable way to do it in front of an audience.
- 6 **Or set it by hand.** **Start time (UTC)** and **End time (UTC)** are both required. Click the small calendar glyph at the right-hand edge of a field to open the picker, or type into the segments. Two things to watch: the field shows the order your *browser* uses (a US-locale Chrome writes mm/dd/yyyy), and the value means UTC, not your local clock. The line underneath always echoes back the exact window it understood — read it before pressing Create.
- 7 **Press Create case.** The case is created and opens on its own map screen.
- 8 **Run the chain.** On the case screen press **Run pipeline**, choose the mode, and confirm. Watch progress under the **Pipeline** tab in the right-hand panel — every step appears with its everyday name, a progress bar and its status.

The two limits, and why they exist

Limit	Reason
Area at most 250,000 km ²	A bigger box means more satellite images and a much slower run, with no gain in precision.
Window at most 30 days	Oil drifts. Running the sea backwards over a longer gap spreads the answer so wide it stops being useful.

Do not have real satellite credentials to hand? There is a repeatable demonstration scenario that generates a complete fake case — fake slick, fake weather, fake ships — so the whole chain can be shown end to end. Everything it produces is labelled **SYNTHETIC** everywhere it appears and can never be mistaken for a real observation.

What happens after you press Run

The steps run in the order shown in section 1, each as its own job. You can leave the screen; the run continues. Completed and failed runs appear in the bell icon in the top bar, and the whole history is in the **Activity feed**. If a

step fails, it says which step and why — a failed data source is always a stated reason, never a silent gap.

5 · Inside a case — the six tabs

Open any case and you get six tabs. They are in the order an investigation actually happens, so reading left to right tells the story.

Walkthrough </cases/<id>/walkthrough>

Start here if anything is unclear. The whole investigation told as eight numbered steps in plain English, each carrying the real figure that step produced in this case, and each with a "Careful:" line saying what that step does *not* prove.

This is the tab to show someone who has never seen the system before.

Map </cases/<id>>

The investigation on one map, with everything switchable.

THE EIGHT LAYERS, EACH WITH ITS OWN ON/OFF SWITCH AND OPACITY SLIDER

Layer	What it is
Area of interest	The box that was drawn when the case was created.
Sentinel-1 scene footprint	The patch the satellite image actually covers.
Origin probability region	Where the oil plausibly started — nested contours, not a point.
Reverse-drift particles	The individual particles pushed backwards through the weather.
Oil probability	The model's raw per-pixel output painted over the map — what it actually saw.
Detected slick	The final outline the detection produced.
Vessel trajectories	The path each ship took through the water.
Vessels	Ship positions. Hover any one for its details.

There is also a **timeline** along the bottom with a **Play** button, so you can watch the ships move through the window, and a right-hand panel with three tabs: **Layers**, **Pipeline** (every step and its status) and **Candidates**.

Spill details </cases/<id>/spill>

The slick itself, and whether it could be something harmless.

- **Four figures** — slick area, detection confidence, verification confidence, and when the image was acquired.
- **Look-alike verification** — a table of the seven checks: Rule, Outcome, Observed, Threshold, Weight, and "What it means" in plain words.

- **Where — and what the model saw** — a **Before · scene / After · model output** toggle over the map.
- **Conditions at acquisition** — the wind the checks were evaluated against — and the full **Run manifest**.

Careful: detection confidence and verification confidence measure two different things and are deliberately never multiplied together. Neither is a probability that any ship discharged anything.

Drift & origin </cases/<id>/drift>

Where the oil plausibly started.

- **Origin confidence** (how tightly the particles bunched), **region area**, the **inferred discharge window**, and the **number of particles**.
- **Probability contours** table — each contour with its probability mass, its area and what it means.
- **Simulation parameters** — engine, duration, time step, ensemble members and the random seed, so the run reproduces exactly.
- **Environmental forcing** — the wind and current variables used, with mean, min and max.

Careful: this is a region, never an exact discharge coordinate, and it is not a claim that a discharge happened. Small errors in wind grow the further back you run.

Vessel ranking </cases/<id>/ranking>

The ships that were inside the origin region during the window.

The table has: **Rank · Candidate vessel · Investigative score · Evidence strength · Closest approach · Evidence**. Press **Show evidence** on any row to open all six factors, their weights, and the arithmetic that produced the total — so every rank can be argued with.

THE SIX FACTORS

- | | |
|---|--|
| • Origin proximity — how close it came | • Heading match |
| • Time match — how well its timing fits | • Speed match |
| • Trajectory match — how well its path fits | • AIS reliability — how complete its own reporting was |

Careful: this ranks who is worth asking about first. It never establishes responsibility, and the nearest ship is not automatically the source. Notice the panels at the top of this screen explaining what the score is, what proximity means, and **why the bands are capped**.

Evidence report </cases/<id>/report>

The full record as one document, rendered by the server exactly as it would be handed over.

Buttons: **Print**, **Download HTML**, **Download PDF**, **Reload**. Below it an **Evidence artifacts** table lists every file the run produced — artifact, media type, size, SHA-256 checksum, data mode and when it was produced.

The checksum matters: it is how someone later proves the file they are holding is the one this investigation actually used.

6 · Intelligence screens

Vessel registry /vessels **EVERYONE**

Every ship your cases have observed in AIS broadcasts, grouped by its radio identity (MMSI).

WHAT IS ON IT

- **Three counters** — distinct vessels observed, ranked as a candidate at least once, and seen in more than one case.
- **A table:** Vessel · Provenance · Observed in · Candidate in · Best rank · Last seen (AIS).
- **Filters:** All · Candidates only · Seen in 2+ cases. Plus a search box (name, MMSI, IMO or call sign), a sort control, and a **Table / Cards** view toggle.

HOW TO USE IT

Search a ship you are curious about, or filter to "Seen in 2+ cases" to find repeat appearances. Click a ship to open its own page with its track, AIS quality and history across your cases.

Two cautions printed on the screen itself. Absence from this list is not absence from the sea — public AIS coverage is incomplete. And appearing in several cases is history, not evidence: a ship on a regular route through a busy lane will reappear simply because it is often there.

Activity feed /activity **EVERYONE**

A running log of everything the system has done, newest first, grouped by day.

Each entry shows the step's everyday name with its machine name beneath — for example **Slick detection** `ml.detect`, **Drift hindcast** `drift.hindcast`. Filter by **All / Running / Completed / Failed**, each with a live count, and narrow further by **Stage**, by **Case**, or with the search box.

Use it when a case seems stuck, or you want to prove what ran and when.

Compare cases /compare **EVERYONE**

Two investigations side by side, so the same measurement can be read across both without flipping between tabs.

HOW TO USE IT

- 1 Pick **Case A** from the first dropdown.
- 2 Pick **Case B** from the second.
- 3 Read down the sections — they line up row for row.

WHAT IT COMPARES

- **Where and when the cases sit relative to each other** — both areas and both time windows.
- **Detections** — slick sizes and confidences.
- **Drift & origin** — origin regions and confidences.

- **Candidate vessels** — the two ranked lists.
- **Vessels seen in both** — a table of Vessel · Type · In case A · In case B.

What it is good for. Two slicks in the same waters a week apart: did the same ships appear in both? The "Vessels seen in both" table answers that in one look. Remember it is a starting question, not a finding.

7 · Administrator screens

These two do not appear in an analyst's sidebar at all. They are about the health of the system, not about any investigation.

System status /admin **ADMINISTRATOR**

Is the machine healthy, and what is it plugged into? Refreshes every 15 seconds; there is also a **Refresh** button.

FIVE SECTIONS

Section	What it tells you
Components	Component · Status · Latency · Detail — the database, Redis queue, object storage, job queue and workers. Anything not UP needs attention.
Data sources	Port · Active implementation · Data mode · Credentials configured · Requires. This is where you see at a glance whether a feed is REAL or SYNTHETIC, and which environment variable a real one needs.
Model versions	Model · Version · Framework · Active · Evaluation — which detection model is live.
Recent jobs and Failed jobs	Queue counters (Queued, Running, Completed 24h, Failed 24h) and every failure with its stated reason.

THE ADMINISTRATOR'S ROUTINE

1. All components UP?
2. Failed jobs in the last 24 hours — zero, or a reason you understand?
3. Data sources in the mode you expect for this demonstration?
4. The intended model marked active?

Credentials are read on the server only and never reach the browser. This screen tells you *whether* a credential is configured, never what it is.

ML Ops /ml-ops **ADMINISTRATOR**

Which detection model is in use, what it scored when it was tested, and how the queue is behaving.

WHAT IS ON IT

- **Registered model versions** — one card per model, with framework, input channels, input size, when it was registered, and its **Evaluation**.
- **Pipeline throughput** — Queued, Running, Completed (24h), Failed (24h).
- **Stage timing** — how long each step takes, per run.
- **Failure reasons** — grouped, so a repeating problem is obvious.
- **How models reach production** — the path from Dataset → Training → Evaluation → Registration → Serving.

Read the evaluation label carefully. A model with no recorded evaluation is shown as *unmeasured* — never as a score of zero. And where numbers come from a validation split rather than an independent test set, the card says so, because the two mean different things.

How to read this [/help](#) **EVERYONE**

The reference for every number on screen: the investigative score, the evidence bands, the origin region, look-alike checks, AIS gaps and coverage, data provenance, the nine safeguards, a glossary and the keyboard shortcuts. It also has a button to replay the introductory tour.

8 · A five-minute walkthrough for visitors

If you have to show this to someone in five minutes, this order works.

- 1 **Start on the public page** (sign out, or open it in a private window). It states the problem and the chain in one screen.
- 2 **Sign in.** Point at the top-right corner: the role is visible. Point at the bottom of the sidebar: it says whether any data in use is synthetic.
- 3 **Dashboard.** Four counters, recent cases. Ten seconds — do not linger.
- 4 **Open a case → Walkthrough tab.** Spend most of your time here. Scroll the eight steps. Each has a real number and a "Careful:" line. This is the whole product in one screen.
- 5 **Stop at step 7 and point at the band.** The top score is high, yet the label reads MODERATE — because the evidence cannot separate the top candidates, so the system holds the label down and prints the reason. **This is the single best thing to show.** Most systems would just show a big number.
- 6 **Jump to the Map tab.** Turn on *Oil probability* to show what the model actually saw, then press **Play** on the timeline to move the ships.
- 7 **Vessel ranking → Show evidence** on the top row. Six factors, weights, arithmetic. Nothing hidden.
- 8 **Finish on the Evidence report.** Sources, checksums, model version, caveats — what would actually be handed to an investigator.

Three questions you will be asked

Question	Answer
"So it catches the ship that did it?"	No, and it does not claim to. It produces a shortlist of ships worth asking about, with the reasoning open to inspection. Deciding responsibility is a job for people with powers this system does not have.
"How accurate is the detection?"	Open ML Ops and read the card. Where the numbers come from a validation split rather than an independent test set, it says so. Nothing is claimed that was not measured.
"What if a ship switches off its transponder?"	A gap is shown as a gap, and it lowers confidence in what can be said about that ship. It never raises a ship's score, because there are many innocent reasons for a gap.

9 · When something looks wrong

What you see	What to do
ML Ops and System status are missing from the sidebar	You are signed in as an analyst. Check the top-right corner and sign in with an administrator account (section 2).
"Administrator access required" on the System status page	The same cause. The data-sources table below the message still works, because knowing whether data is synthetic is not an administrative privilege.
A case sits at RUNNING and does not move	Open the Pipeline tab on the case, or the Activity feed, and look for the step that has not completed. A failed step always states its reason.
The map shows a plain grid instead of satellite imagery	The imagery tiles could not be reached, so the map fell back to its offline basemap and said so. Every evidence layer still works. Use the basemap switcher to try again.
A number shows as a dash	That value is unknown, and is deliberately not shown as zero. "We could not measure this" and "this was zero" are different facts.
A score looks high but the band says MODERATE	Working as intended. The evidence cannot separate the top candidates, so the label is capped and the reason is printed next to it.
You want the explanation panels back after hiding them	Click the small "What am I looking at?" chip near the top of any screen.

Glossary

Term	Meaning
SAR / radar imaging	A satellite that sends its own pulses and listens for the echo, so it works at night and through cloud. Rough sea looks bright; smooth sea looks dark.
Sentinel-1	The free European radar satellites that supply every image here.
Slick	A thin film of oil on the surface. It damps the ripples underneath, which is what radar sees.
Look-alike	Something that is not oil but looks identical on radar: low wind, algae, a rain cell, river water.
AIS	The public radio broadcast carrying a ship's identity, position, speed and heading.
MMSI	The nine-digit radio identity a ship broadcasts. Everything here is grouped by it.
IMO number	A permanent seven-digit number for a hull. It survives renaming, but is not always broadcast.
Origin probability region	The area the oil plausibly came from, as nested contours — never an exact point.
Investigative score	Six weighted factors combined into one number from 0 to 1, used only to sort candidates.
Evidence band	The LOW / MODERATE / HIGH label. Decided by the server, and capped when candidates cannot be separated.

Provenance	REAL, SYNTHETIC or MIXED — where a result's inputs came from.
Area of interest (AOI)	The patch of sea a case covers, drawn when it was created.

If you remember one sentence from this guide: SPILLTRACE narrows an ocean down to a short list of ships worth asking about, and shows its working at every step — but it never establishes who is responsible, and it is built so that it cannot pretend otherwise.